Trevor Duong

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EDUCATION

University of California, San Diego **La Jolla, CA**
B.S. Computer Science (GPA: 3.76/4.0, Regents Scholarship Recipient) *Graduation Date: June 2027*

- **Relevant Coursework:** Deep Learning | Natural Language Processing | Machine Learning Algorithms | Data Structures and Algorithms | Databases | Networked Services | Operating Systems | Computer Security

PROFESSIONAL EXPERIENCE

Sked AI **San Francisco, CA**
Co-founder, Backend Lead *May 2024 - Present*

- Leading backend engineering at an **early-stage AI startup (Berkeley SkyDeck + F6S)** building a smart scheduling assistant that interprets natural-language constraints and computes optimized schedules across mobile and web platforms.
- Building and load-testing a **cloud-native microservices backend** using **Supabase + PostgreSQL (SQL/NoSQL)** deployed on **AWS EC2**, validating scalability and fault tolerance through load testing and automated **CI/CD (GitHub Actions)** pipelines.
- Architecting **LangChain-based NLP pipelines** in **Python** to convert natural language prompts into structured task and location constraints, using **agentic AI techniques** such as conditional tool use and planning to associate tasks with real-world locations and automate scheduling with minimal input.
- Developing a **multi-layer optimization pipeline** with **Google OR-Tools**, applying **algorithmic problem-solving** and **data structure refinement** to minimize overhead and **scale constraint-solving performance** for high-load scheduling workloads.
- Leading a **security audit** of the multi-tenant backend, identifying and remediating **9 critical and high-severity vulnerabilities** (including cross-tenant IDOR issues, row-level security gaps, and an XSS flaw in the LLM chat interface) through 4 production migrations and a documented audit playbook.
- Directing backend technical strategy for a **4-engineer team**, including code review standards and **Agile** practices.

UC San Diego Computer Science & Engineering **La Jolla, CA**
Instructional Assistant / Undergraduate Tutor (CSE 151B: Deep Learning) *June 2026 - Present*

- Supporting students in class and during office hours with deep learning concepts and assignments, including backpropagation, loss function design, and modern architectures (MLPs, CNNs, RNNs, Transformers).
- Debugging **PyTorch** implementations with students for programming assignments and Kaggle competitions; creating and testing the assignments and exam material for the course using **PrairieLearn Automation**.

PROJECT

LLM Mathematical Reasoning Capstone (PyTorch, Hugging Face TRL, GRPO, Kubernetes) **La Jolla, CA**
CSE 151B Deep Learning Capstone *April 2026 - June 2026*

- Improved a fixed 4-billion-parameter LLM's (Qwen3-4B) held-out mathematical reasoning accuracy from **0.407 to 0.581** (a 43% relative gain) under a no-additional-compute constraint, isolating all gains to method rather than scale.
- Ran approximately **40 version-controlled experiments** spanning supervised fine-tuning, **parameter-efficient methods (LoRA, QLoRA), and GRPO reinforcement learning**, including diagnosing and fixing output truncation as the primary failure mode.
- Trained on UCSD's DSMLP **Kubernetes cluster** with checkpoint-to-**Hugging-Face-Hub** resumption across a 12-hour container ceiling, and used **vLLM** continuous batching to cut evaluation time from about **93 hours to 2 to 4 hours**.

Distributed Video Streaming Platform (Go, gRPC, AWS EC2) – TDStream **La Jolla, CA**
Distributed Systems Project *April 2025 - June 2025*

- Developed a **full-stack video streaming platform in Go**, supporting adaptive playback via **FFmpeg** and **RESTful APIs**, improving video delivery speed and stream quality for end users.
- Built a **distributed storage system** using **consistent hashing** and **gRPC** (Protocol Buffers over HTTP/2 on TCP), enabling real-time video distribution across a dynamically changing set of storage nodes with fault tolerance and horizontal scalability.

Adversarial Prompt Injection Classifier (PyTorch, Hugging Face) **La Jolla, CA**
Machine Learning Project *January 2026 - March 2026*

- Engineered an end-to-end **NLP classification pipeline** using **PyTorch** and **Hugging Face** to detect adversarial jailbreak attempts and secure LLM interactions.
- Applied **Parameter-Efficient Fine-Tuning (PEFT)** techniques to a pre-trained **DeBERTa** transformer, optimizing model weights for high-accuracy inference and achieving a **96% F1-score** on a held-out test set of zero-day attacks.

SKILLS

Languages: Python, Go, Java, C++, C, SQL, JavaScript, HTML/CSS, Bash

AI & ML: PyTorch, Hugging Face (Transformers/TRL), Reinforcement Learning (GRPO), PEFT/LoRA/QLoRA, LLM Applications, AI Agents, LangChain, Prompt Engineering, NLP Pipelines, Model Evaluation

Backend & Systems: Microservices, REST APIs, Distributed Systems, Workflow Automation

Databases & Cloud: PostgreSQL, Supabase, SQLite, AWS (EC2, S3), Docker